Model 2724A/CPR

Programmable Precision Resistance Standard

Operation Manual

VALHALLA SCIENTIFIC

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SECTION I - UNPACKING AND INSTALLATION

1.1 UNPACKING AND INSPECTION

If the shipping carton is damaged, request that the carrier's agent be present when the unit is unpacked. If, when unpacked, the unit shows external damage, the carrier's agent should authorize repairs before the unit is returned to the factory. If the shipping carton is damaged, the unit may have suffered internal damage in transit that may not be evident until the unit is operated or tested to verify conformance with its specifications. If the unit fails to perform or fails to meet the performance specifications of Section II, notify the carrier's agent and the nearest Valhalla Sales Office. Retain the shipping carton for the carrier's inspection. DO NOT return equipment to Valhalla Scientific or any of its sales offices prior to obtaining authorization to do so.

1.2 INITIAL ADJUSTMENTS

The only adjustments required before placing the unit in operation are to set the rear panel switch to the local power voltage and verify that the proper fuse is installed as listed in Table 1-1.

Table 1-1. Voltage Switch Settings and Fuse Ratings.

SUPPLY VOLTAGE	SWITCH SETTING	FUSE RATING
104 - 127	115	0.5 Amp Slo Blo
208 - 254	230	0.2 Amp Slo Blo

1.3 INSTRUCTIONS FOR BENCH USE

The unit is delivered with all required hardware for bench use installed and special instructions for use in that manner are not necessary. However, before connecting the unit to the local power source, verify that the power cord is equipped with a three-terminal connector (see the Safety Instructions of paragraph 1.5).

1.4 INSTRUCTIONS FOR RACK MOUNTING

Optional brackets are available for mounting the Model 2724A in an equipment rack. These are listed in Section VII. The size of the unit and the location of its center of gravity dictate that it should be supported on both sides along its entire length through the use of "trays" or slides. If it is to be transported while mounted in a rack, it should be supported such that movement in any axis is not possible.

Note that the specifications listed in Section II indicate that accuracy of the unit is degraded at abnormally high temperatures. Therefore, it is recommended that "spacer" panels at least 1.75 inches high be installed

between this and any other units in the rack to insure freedom of air flow. Under no circumstances should the ambient air temperature around the unit exceed 50°C while the unit is in operation or 70°C when power is removed.

1.5 SAFETY PRECAUTIONS

The power plug should be a three-contact device and should be inserted only in a three-contact mating socket where the third contact provides a ground connection. Where power is provided through an extension cable with several outlets, the ground connection must be continuous. Any discontinuity in the ground lead may render the unit unsafe for use.

SECTION II - SPECIFICATIONS

The operating specifications for the 2724A are listed in table 2-2.

The environmental specifications are listed below in Table 2-1.

Table 2-1: General Specifications

Dimensions:	89mm (3.5 in) H x 432mm (17 in) W x 432mm (17 in) D
Weight:	7.2kg (16 lb) net, 10.5kg (23 lb) shipping
Power:	115/230 VAC ± 10% at 48 to 440 Hz, 30 VA
Temperature:	0 to 50C - Operating -30 to 70C - Storage
Humidity:	5 to 70% RH (non-condensing)

Table 2-2: Operating Specifications

Accuracies below are relative to NBS traceable standards and are valid at +5°C from calibration temperature within 15°C to 30°C. Add 21ppm for absolute accuracy utilizing Valhalla's production calibration traceability uncertainties.

2724A Active Mode Operation

Resistance in Ohms	Current		Accuracy (for 90 Days) ±5°C)	Stability (DC to 1Hz)		Temperature Coefficient in ppm/°C	Setting Time in seconds	
	Min.	Max.		24 Hr.	1 Year		Change in Current	Change in Value
0 to 120	500µA	120mA	7ppm+2mΩ	2ppm	10ppm	1.5	2	2
120 to 1.2K	50µA	12mA	7ppm+7mΩ	2ppm	10ppm	1.5	2	2
1.2K to 12K	5µA	1.2mA	7ppm+50mΩ	2ppm	10ppm	1.5	2	2
12K to 120K	500nA	120µA	7ppm+500mΩ	2ppm	10ppm	1.5	2	2
120K to 1.20M	50nA	12µA	12ppm+5Ω	2ppm	10ppm	3	2	2
1.2M to 12M	5nA	1.2µA	20ppm+50Ω	2ppm	15ppm	5	3	2
12M to 120M	500pA	120nA	40ppm+1KΩ	500Ω	50ppm	15	4	2
120M to 1.2G	50pA	12nA	0.1%+50KΩ	50KΩ	0.05%	15	6	3
1.2G to 11G	5pA	1.2nA	0.1%+5MΩ	5MΩ	0.05%	15	15	5

Table 2-2: Operating Specifications (Continued)

2724A Fast Mode Operation (Active)

Resistance in Ohms	Current pk. max.	One Year ±5°C Absolute Accuracy in Ohms	Temperature Coefficient in Ohms/°C	Setting Time in milliseconds		0.05% Error Fre-quency
				Change in Current	Change in Value	
0 to 120	120mA	0.04	0.006	0.1	5	3KHz
120 to 1.2K	12mA	0.4	0.06	0.1	5	3KHz
1.2K to 12K	1.2mA	4	0.6	0.1	5	3KHz
12K to 120K	120µA	40	6	0.2	5	2KHz
120K to 1.2M	12µA	400	60	1	5	500Hz
1.2M to 12M	1.2µA	6K	600	10	10	50Hz
12M to 120M	120nA	60K	6K	500	100	--
120M to 1.2G	12nA	600K	60K	5 Sec.	2 Sec.	--
1.2G to 11G	1.2nA	6M	600K	15 Sec.	5 Sec.	--

Table 2-2: Operating Specifications (Continued)

2724A CPR Mode Operation (Slow or Fast Modes)

Resistance in Ohms	Current Max.	Accuracy (for 90 days) ±5°C	Stability (DC to 1Hz)		Temper-ature Coeff-icient in ppm/°C	Settling Time in Milliseconds	
			24 Hr.	1 Year		Fast Mode	Slow Mode
100	120mA	7ppm+2mΩ	2ppm	10ppm	1.5	0.1	100
1K	12mA	7ppm+7mΩ	2ppm	10ppm	1.5	0.1	100
10K	1.2mA	7ppm+50mΩ	2ppm	10ppm	1.5	0.1	100
100K	120µA	7ppm+500mΩ	2ppm	10ppm	1.5	0.1	100
1M	12µA	12ppm+5Ω	2ppm	10ppm	3	0.3	1 Sec.
10M	1.2µA	20ppm+50Ω	2ppm	15ppm	5	3	5 Sec.

Output Configuration: 2-wire/4-wire, front and rear terminals standard.

Leakage Current: ± 2pA ± 0.2pA/°C

Power Coefficient: 0.15ppm/mW

Noise and Thermals: (DC to 10Hz, 4-terminal)
CPR Mode ± 2uV Maximum
Standard Mode ± 4uV Maximum
Fast Mode ± 30uV Maximum

Maximum Input: 25V peak or 120mA (if option CPR installed: 15mA maximum in active modes)

Warm-up Time: 15 minutes to within 5ppm of final value

SECTION III - MANUAL OPERATION

3.1 GENERAL

The Model 2724A is equipped with front panel controls listed in Tables 3-1(a), (b) and (c).

A more complete description of key functions is provided in the following paragraphs.

Upon initial application of power, the unit performs various self-check procedures resulting in possible fault messages. If the self-check is satisfactory, the display will greet the operator with "HI THERE". ("HELLO", "GUTEN TAG", "BONJOUR" or a greeting message in another language may appear on units shipped outside the United States). Approximately one-second later either "2724A V-FB" or 2724CPR V-FB" will be displayed. Approximately one-second later, "CAL DATA OK" will appear in the display. If the self-check determines that a fault exists, an error message will appear in the display. Some of the messages are listed in Table 3-2. If the unit has the IEEE option installed, the address will be displayed for one second.

3.2 CONNECTIONS

The Model 2724A is a four-wire instrument. It is equipped with separate terminals for connection of the current source and voltage measurement leads of the measuring instrument. However, it may be used as a two-wire instrument as well. Directions for use in both modes are included in following paragraphs.

CAUTION

The 2724A provides rear panel connectors wired in parallel with those on the front panel. Under no circumstances should connections to the front and rear panels exist at the same time.

3.2.1 FOUR-WIRE CONNECTIONS

When the Model 2724A is used with a four-wire measuring instrument, the current source leads of the measuring instrument should be connected to the "I" terminals of the 2724A. The voltage measuring leads should be connected to the "V" terminals. The 2724A can accommodate either polarity. However, care should be exercised to ensure that both leads connected to a pair of like colored terminals on the 2724A are of the same polarity.

Table 3-1(a). Front Panel Controls (2724A and 2724CPR)

MARKING	FUNCTION
ON/OFF	Selects/deselects digit step mode (See 3.4)
←	Shifts controlled decade up one decade (See 3.4)
→	Shifts controlled decade down one decade (See 3.4)
↑	Increments selected decade (See 3.4)
↓	Decrements selected decade (See 3.4)
Ø to 9 and .	Used for numerical entry of data (See 3.3)
CLR	Resets displayed value (during entry) (See 3.3)
MAN	Selects manual control (if enabled) (See 4.3)
STO MEM	Stores present value in one of 1Ø memories (See 3.5)
RCL MEM	Recalls value stored in one of 1Ø memories (See 3.5)
RCL LAST	Recalls last value (See 3.5)
Ω	Pressed at end of entry to indicate unit ohms (See 3.3)
KΩ	Pressed at end of entry to indicate kilohms (See 3.3)
MΩ	Pressed at end of entry to indicate megohms (See 3.3)
IEEE ADDR	Allows the IEEE unit address to be displayed or changed (See 4.3)

Table 3-1 (b)

Front Panel Controls (2724A Without Option CPR)

MARKING	FUNCTION
2 WIRE	Selects 2-wire offset compensation (See 3.6.1)
4 WIRE	Selects for no 2-wire compensation (See 3.6.1)
SLOW MODE	Selects for "Slow mode" operation (See 3.7)
FAST MODE	Selects for "Fast mode" operation (See 3.7)

Table 3-1 (c)

Front Panel Controls (2724A with Option CPR)

MARKING	FUNCTION
2 WIRE	Alternately (de)selects 2-wire compensation (See 3.6.2 and 3.8)
FAST MODE	Alternately selects Fast or Slow modes (See 3.9)
CARD PNT	Alternately selects CPR or Active modes (See 3.10)
PPM DEV	Allows for computation of UUT error (See 3.11)

Table 3-2. Error Message Displays.

MESSAGE	MEANING
All decimal points illuminated	ROM failure
All horizontal middle elements illuminated.	RAM failure
OUT OF CAL	Component out of tolerance
CAL DATA BAD	Data corrupted (requires calibration)
CLOCK FAILURE	Faulty clock circuit
MATH OVERFLOW	Contact Valhalla Service Office if this occurs
CANNOT CONTROL	See maintenance Section VII
MEMORY DATA BAD	Memory data corrupted
SETTLING	This is not an error message, but an indication that the instrument has not stabilized since new data was entered.
ATOD TIMEOUT	If this message persists or flashes then refer to maintenance Section VII.

3.2.2 TWO-WIRE CONNECTIONS

When the measuring instrument has only two wires, they should be connected to the "I" terminals of the 2724A. In such cases, to minimize measurement errors, the leads should be as short as possible, especially if the resistance to be measured is in the order of a few ohms. The two-wire mode should be selected to compensate for zero errors.

3.3 NUMERIC VALUE ENTRY

To select a resistance value for the output of the Model 2724A, the keys marked with "Ø" through "9", ".", "CLR", "Ω", "KΩ", and "MΩ" are used. The following examples illustrate the procedure:

To select 11.458 Ohms Press 1 1 . 4 5 8 Ω

To select 1Ø5 M Ohms Press 1 Ø 5 MΩ

As the numbers are entered, they appear in the display. Pressing the "Ω", "KΩ" or "MΩ" key terminates the entry. Should an error be made during entry and before the "Ω", "KΩ" or "MΩ" keys are pressed, pressing the CLR key will return the display to zero. Resistance values may be entered in any form. For example, 9ØØ Ω, Ø.9KΩ , and Ø.ØØØ9 MΩ are valid entries for 9ØØ ohms.

3.4 STEP CONTROL

Any digit of a previously completed entry may be altered using the step control feature which is selected with the STEP CONTROL ON/OFF key. The least significant digit is initially selected for alteration. However, the keys with the left and right arrows enable the selection of other digits for alteration without affecting any of the other digits except for carry or borrow. The selected digit flashes. The keys with the up and down arrows increment and decrement the digit. If a cardinal point (Ø) is traversed, the unit automatically maintains the same digit for control, but alters higher or lower digits as necessary.

There are some points to remember:

1. The STEP CONTROL ON/OFF control is used to enter and exit the step control mode. The indicator in the key is illuminated when the mode is active.

2. The minimum resistance is Ø ohms. Attempting to step below zero will cause Ø ohms to be selected. Similarly, the maximum resistance is 1Ø.99999G Ohms.

3. Numeric entry will abort step-control and vice-versa.

3.5 MEMORIES

The Model 2724A has two types of user controllable memories:

1. Last Value Memory -- The unit automatically stores the last entered value, which may be recalled when not in the step control mode by pressing the RCL LAST key. A very useful feature of this control is that it exchanges the last and present values. Thus, if two resistance values need to be repetitively exchanged, pressing the RCL LAST (RECALL LAST) key will cause the unit to alternate between the two values. Note that only numerical values are stored.

2. User Definable Memory -- Any value presently in the display may be stored by pressing the STO MEM (STORE MEMORY) key and the appropriate digit key. The digit key selects one of ten memories (Ø through 9) in which the data will be stored. The contents of the memories may be recalled at any time. Press the RCL MEM key and the digit key that corresponds to the memory from which data is to be recalled.

Some points to remember:

1. The data is stored in non-volatile memories. Therefore, the data stored prior to shutdown will be in memory when power is again

applied. The non-volatile memory is useful in ATE applications where the same values may be used over a period of days or weeks.

2. The value previously stored in memory location Ø is used as the initial value upon application of power.

3.6 TWO-WIRE/FOUR-WIRE SELECTION

3.6.1 2724A WITHOUT CPR

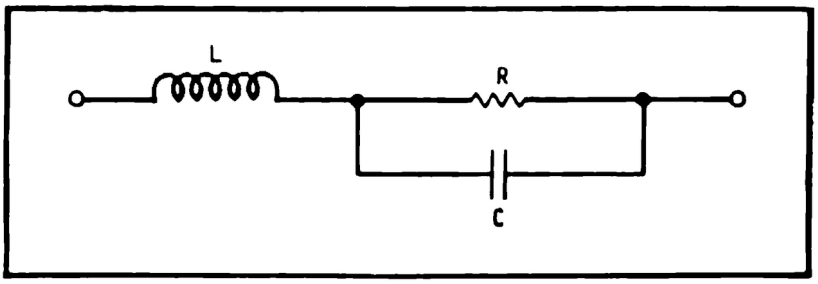
The two keys marked 2 WIRE and 4 WIRE are used (the adjacent LED indicating which mode is presently active).

In two-wire mode the unit actually outputs the displayed value less the calibrated two-wire offset.

3.6.2 2724A WITH CPR

The 2-WIRE key alternately switches between two-wire (LED on) and four-wire (LED off) modes. In two-wire active mode the unit outputs the displayed value less the calibrated two-wire offset. Whilst in CPR mode the displayed value is modified by the two-wire offset.

Figure 3-1. Slow Mode Equivalent Circuit



R = required resistance C = 50 R_{ref}/R nanofarads

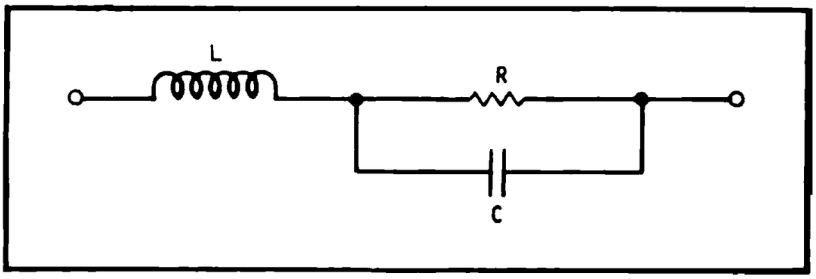
L = R millihenries (R is in ohms)
(R is in ohms) (R_{ref} is the full-scale value in ohms)

3.7 SLOW-MODE AND FAST-MODE (2724A WITHOUT CPR)

When Slow-mode is selected (adjacent LED on) the unit operates as a high-precision DC resistor with an equivalent circuit shown in Figure 3-1. In this mode the unit requires a stable DC current for correct operation.

When Fast-mode is selected (adjacent LED on) the unit operates as a medium-precision DC/AC resistor with an equivalent circuit shown below:

Figure 3-2. Fast-mode Equivalent Circuit



Where R is the required value
C is given by the table below:

0 - 120 Ω	50000pF nominal
0.12 - 1.2K Ω	5000pF nominal
1.2 - 12KΩ	500pF nominal
12 - 120KΩ	60pF nominal
0.12 - 1.2M Ω	50pF nominal
1.2 - 12M Ω	50pF nominal
12 - 120M Ω	1000pF nominal
0.12 - 1.2GΩ	1000pF nominal
1.2 - 11G Ω	1000pF nominal

L is ± 30 R_{ref} nH max . (R_{ref} in ohms)

When measured using a two-terminal hook-up the capacitor "C" above appears in series with 10KΩ.

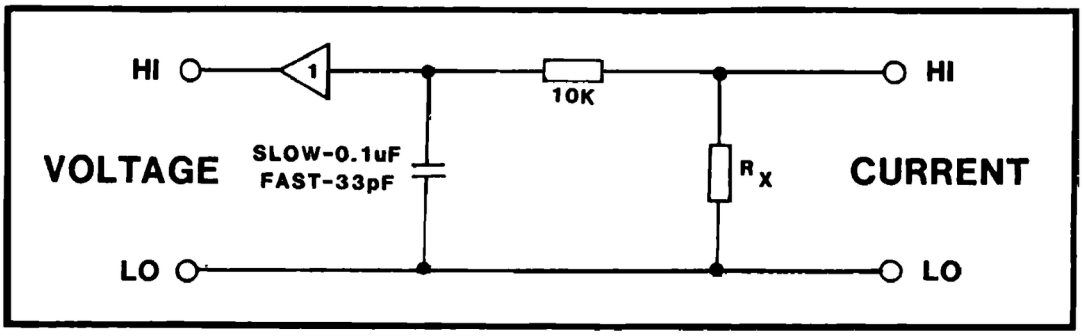
3.8 TWO-WIRE KEY (2724A WITH CPR)

This key alternately selects TWO WIRE and FOUR WIRE modes of operation. The adjacent LED is on whenever TWO WIRE is selected, and the units' output is adjusted to compensate for the two-terminal zero error entered during calibration (in ACTIVE mode) or the displayed value is adjusted to include the two-terminal zero error (in CPR mode).

3.9 FAST-MODE KEY (2724A WITH CPR)

This key alternately selects FAST and SLOW modes. The adjacent LED is on while in FAST-mode. All of the information regarding the equivalent circuit in the 2724A (3.7 above) is applicable when in ACTIVE mode. In CPR mode, the equivalent circuit is as shown below:

Figure 3-3. CPR Mode Equivalent Circuit



3.10 CARD PNT KEY

This key alternately selects CPR and ACTIVE modes of operation. Whilst in CPR mode the adjacent LED is on.

In ACTIVE mode the unit is just as 2724A (except the maximum current is 15mA).

In CARD PNT mode (CPR) the unit selects the cardinal point value for the "range" in which the entered value falls, (e.g. 8K will call up for 10KΩ, ØΩ will call up for 100Ω) up to a maximum value of 10M . When in CPR mode the Step Control function is disabled, the displayed value being the calibrated value of the resistor (modified for two-terminal zero offset if required).

3.11 PPM DEV KEY

This key selects for the next entered value to be compared with the currently displayed CPR Value and the deviation (in parts per million) to be displayed for 4 seconds. This key is only functional in CPR mode, and the adjacent LED is on whilst the unit is awaiting the entered value. Whilst the LED is on, this condition may be cleared by pressing "PPM DEV", or exiting CPR mode using the "CARD PNT" key.

The user should note that the maximum displayable error is +200000/-1999999 ppm.

SECTION IV - IEEE-488 INTERFACE OPERATION

4.1 IEEE-488, AN EXPLANATION OF THE BUS

The reader that is not familiar with the requirements of IEEE Standard 488 (1981) (IEEE-488) will want to review the following paragraphs which provide explanations of terms, commands and some examples of commands. For a complete explanation of the requirements, refer to the IEEE-488 Standard.

4.2 DEFINITIONS

The following are definitions of the terms used in describing the IEEE-488 interface.

Bi-Directional Bus: A 'highway' used for two-way communication, both input and output data being conveyed on the same lines.

Bus: A data link which is usually a set of several parallel wires within a multi-wire cable.

Bit Parallel: Used to describe data which is present simultaneously on a number of wires in a bus.

Byte: A group of data bits (usually 8) which are treated as a single item of data.

Byte Serial: Information, in bit-parallel bytes, transferred sequentially between devices.

Device Dependent Message: A message containing commands/data specifically for the recipient device.

Handshake: A sequence of signals, controlling the transfer of data over a bus. Each condition, in turn, must occur before the next.

Interface: The part of an instrument or system which enables it to be connected to another via a bus.

Interface Message: A message intended for interface management.

Local Operation: Operation of a device by its front panel controls (also referred to as Manual Control).

Remote Operation: Operation of device under the control of another via a bus.

The IEEE-488 Interfacing Standard (also known as IEC DTC66-(WG3), ANSI MC1-1, GP-IB, HP-IB, etc.) defines a bi-directional bus for interconnecting programmable instrumentation in a bit-parallel, byte serial fashion. It defines limitations as follows:

A maximum of 15 devices may be interconnected by a single bus.

The total bus length may not exceed 20 meters, or the number of devices multiplied by two, whichever is the shorter.

Maximum transmission rate is 1 megabyte per second.

All bus data is digital.

Of the devices on the bus, only one may be a controller, exercising control over all other devices and also capable of operating as a "talker" or "listener". The other devices may be listeners (only able to receive data) or talkers (only able to transmit data) or both. (The 2724A is capable of talking and listening). The controller may address other devices and command them to listen or talk (only one may talk at any one time).

The interconnecting cable contains sixteen signal wires and eight ground returns linking devices into a complete system. The cable connectors consist of a plug/socket combination in order that "daisy-chaining" of units may be achieved. The sixteen signal wires comprise:

Eight data wires (DIO-Ø through DIO-7)

Five management wires (ATN, EOI, SRQ, IFC and REN)

Three "handshake" wires (DAV, NRFD and NADC)

It should be noted that these wires use "inverse logic". That is to say that a low level indicates the "true" (active) state and a high level indicates a "false" (non-active) state.

ATN - Asserted by the controller whenever an address or a command is present on the bus.

EOI - With ATN true, EOI indicates that the controller is polling devices. With ATN false, EOI is driven by the talker to indicate the end of data.

SRQ - Driven by any device. This indicates that a device requires attention (e.g., a fault has occurred). Normally, the controller will respond by polling to determine which device requires service.

IFC - Driven by the controller. This signal, when true, initializes the bus to a reset state.

REN - Driven by the controller. This signal, when true, enables the device to enter the remote state.

The transfer of data on the bus is controlled by the three handshake wires which are arranged to run at the speed of the slowest device on the bus. The talker first waits for all devices to be ready to accept data (checks that NRFD is false) then puts the data on the bus with DAV true. It then waits for all devices to indicate that the data has been accepted, by waiting for DAC to become true, before starting to transfer the next byte of data.

Data is sent in 8-bit bytes on the DIO wires, usually (as in the 2724A) using the ISO-7 standard ASCII characters. Table 4-1 lists each ASCII character and the bus messages applicable to each. Note that the table is divided into two main groups. The primary command group and the secondary command group. The secondary command group is not utilized in the 2724A.

The primary command group is further divided into four sub-groups as follows:

1. The addressed Command Group - applied only to addressed devices.

2. The Universal Command Group - applied to all devices.

3. The Listen Address Group set of device listen addresses.

4. The Talk Address Group - set of device talk addresses.

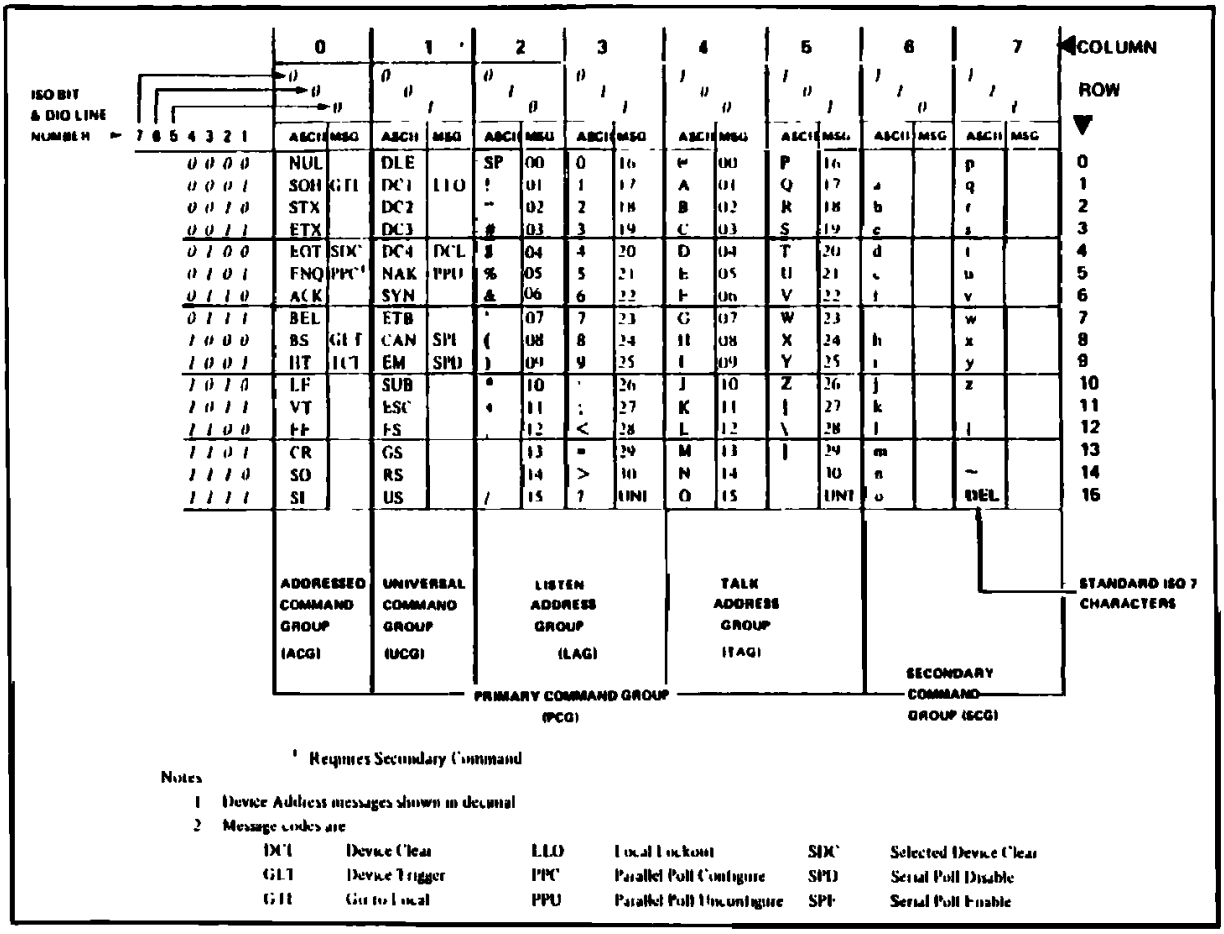
Data in the above command groups is sent with ATN true. When data is sent with ATN false, then it is "device dependent".

4.3 GENERAL

All IEEE-488 device dependent activity with the 2724A takes place via input and output buffers.

Input Buffer - Incoming data is placed in the input buffer as it is received. It is not acted upon until any one of the valid input delimiters is received. Then the commands are decoded and the input buffer contents erased. Input data is held off until the previous data is decoded and actioned. Should an invalid character be found in the input buffer, then the buffer is only decoded up to the error, the rest being discarded. An SRQ can be asserted, if required, should this occur.

Table 4-1. ASCII Character Codes.



The valid input delimiters are:

CR (Carriage Return) or

EOI with the last character.

Output Buffer - With every display update (except fault messages), the output buffer is filled with the contents of the display. This data may be read by the controller any number of times.

The address of the 2724A is preset at the factory at Ø9. However, it may be changed by the user through the front panel keyboard. To enter a new address, press the IEEE/ADDR key and the CLR key to clear the original address. Then press the number keys corresponding to the new address and terminate the entry by pressing the Ω key.

At power-up, the 2724A will be in the local mode (i.e., the unit will respond to commands entered with the front panel keyboard). In order for the 2724A to enter the remote mode, the following sequence, stipulated by IEEE-488, must occur:

1. The remote enable, REN, wire on the bus must be true (pulled low).

2. The 2724A must receive its listen address.

With an HP 85 computer, this is achieved by the statement REMOTE 7Ø9, assuming that the 2724A address is Ø9. In all future examples, an HP 85 computer is assumed as is the 2724A address of Ø9.

When the 2724A is in remote, the REMOTE IEEE-488 indicator on the front panel will be illuminated and all front panel keys (except POWER and MAN) are inhibited. The MAN key, if pressed, will return control to the front panel. If the MAN control is also to be inhibited, then LOCAL LOCKOUT must be sent over the bus. This is achieved with the HP 85 by LOCAL LOCKOUT 7.

The computer may reset the 2724A to the local state at any time by the command LOCAL 7 (or LOCAL 7Ø9 if only the 2724A is to be returned to the local mode). Note also, that a device clear command will return the unit to local control.

4.4 SIMPLE COMMANDS AND EXAMPLES

Six simple commands, with examples, are shown in the following paragraphs.

1. **Setting a Resistance Value** -- To set the 2724A to a numeric resistance value, all that is required is to send the resistance value in ohms. Any form of the number is accepted (exponent, etc.) with a maximum of seven digits.

Digits in excess of seven are ignored. The data will not be processed until terminated or until an exponent character is received.

The delimiter for the exponent may be an ASCII "E" or "e". The overall delimiter is any invalid character within numeric entry.

The following is an example of such a program using the HP 85:

```
1Ø REMOTE 7Ø9
2Ø INPUT A
3Ø OUTPUT 7Ø9 ; A
4Ø GOTO 2Ø
```

This program will prompt a user input of value on the HP 85 and send this value to the 2724A. (It is assumed that the 2724A was just powered-up and the calibrate switch was off).

2. **Step Control** -- To utilize the 2724A's step control facility via the IEEE-488 bus, the user enters the same data as he would from the front panel. The front panel keys are replaced by the following commands:

   DON - Turns step control mode on
   DOFF - Turns step control off

   (Note use of separate commands)

   U - Same as UP arrow
   D - Same as DOWN arrow
   L - Same as LEFT arrow
   R - Same as RIGHT arrow

   See subsection 3.4 for details.

Following is an example of a program of step control using the HP 85:

```
1Ø REMOTE 7Ø9
2Ø OUTPUT 7Ø9 ; "DONLLUURDDOFF"
```

This program will cause the second from the least significant digit to be incremented twice and the next to least significant digit to be decremented. The last part of the program turns the step control mode off.

3. **Reset** -- The 2724A may be reset by either of two methods. First, an IEEE-488 Device Clear will reset the 2724A. Second, the 2724A has a device dependent command A which will produce a reset. This is of particular value when using a controller that does not have a Device Clear capability (such as the Commodore PET).

4. **CPR/Active Mode Selection (Option CPR Only)** -- To select one of these modes the user sends one of the following commands -

CØ - Selects Active Mode (power-up condition)
or C1 - Selects CPR mode.

Example: OUTPUT7Ø9 ; "C1" - Select CPR mode

5. **Two/Four-Wire Selection** -- To select either two- or four-wire correction the user sends one of the following commands -

TØ - Selects four-wire (power-up condition)
or T1 - Selects two-wire

Example: OUTPUT7Ø9 ; "T1" - selects two-wire

6. **Fast/Slow Mode Selection** -- To select either slow- or fast-mode operation the user sends one of the following commands -

MØ - Selects slow-mode (power-up condition)
or M1 - Selects fast-mode

Example: OUTPUT7Ø9 ; "MØ" - Selects slow-mode

4.5 ADVANCED COMMANDS AND EXAMPLES

One advanced command and example of execution is contained in the following paragraphs.

1. **Output Delimiters** - As mentioned earlier, the 2724A updates its output buffer with the display data. The format of this data is as follows:

"nnnnnn KOHMS QnEnPnMnTnfcou" (2724A)
or
"nnnnnn KOHMS QnEnPnMnTnCnfcou" (CPR)

followed by the required output delimiter where:

n is the required number (e.g., En returns the present output delimiter select code)

f is a space if not in step control mode, F if it is.

c is a space if not in calibration mode, C if it is.

o is a space if not in overcurrent, O if it is.

u is a space if not in undercurrent, U if it is.

The required output delimiter is set by sending an "E" followed by the required code (see controller manual for details regarding the required delimiter). The list of available codes is given below:

E0 - Carriage return line feed at end of data

E1 - Carriage return (line feed with EOI asserted) at end of data

E2 - Carriage return only at end of data

E3 - Carriage return (with EOI asserted) at end of data

E4 - EOI asserted with last character

Note that most controllers use E0. However, the Commodore PET should use E3.

Example - If the 2724A was outputting 10.00012M Ohms and was in step control mode and in undercurrent:

```
10 REMOTE 709
20 DIM A$[100]
30 ENTER 709 ; A$
40 PRINT A$
```

The resultant print-out would be:

```
"10.00012 MOHMS Q0E0P0M0T1   U"
```

4.6 SRQ, SERIAL POLL AND PARALLEL POLL

1. **Parallel Poll Response** -- The 2724A may have its parallel poll response bit set by means of the "P" command. The user must send "Pn" to the unit in order to configure the response, where "n" defines the DIO line that is to be asserted by the 2724A. Sending "P0" selects that no response shall be made to a parallel poll.

2. **SRQ Definition** -- The 2724A may assert the SRQ wire for one of several reasons. This is defined by the user sending "Qn", where n is the numeric code for the required reasons.

   Q0 - Disables the 2724A from asserting SRQ

   Q1 - Enables the 2724A to assert SRQ on internal errors, i.e., any displayed fault, overcurrent, undercurrent, unsettled, etc.

   Q2 - Enable to assert SRQ on input data errors, e.g., if "B" is received on the data bus (an un-decodable command)

   Q3 - Enable to assert SRQ on all reasons given in Q1 and Q2.

Q4 - Enable to assert SRQ when settled to a new value.
Q5 - Enable to assert SRQ on all reasons given in Q1 and Q4.
Q6 - Enable to assert SRQ on all reasons given in Q2 and Q4.
Q7 - Enable to assert SRQ on all reasons.

3. **Serial Poll Response Byte** -- When the controller in the system conducts a serial poll, the 2724A responds with a byte of data informing the controller of the reason for the SRQ. If the 2724A did not generate the SRQ, then a zero byte is used as the response. The bit patterns used by the 2724A are listed in Table 4-2.

Table 4-2. SRQ Bit Patterns.

REASON	BIT PATTERN IN RESPONSE
Clock Fault	0100 0001
Math Overflow	0100 0010
Bad Calibration Data	0100 0011
Bad Memory Data	0100 0100
Settled	0101 0000
Calibration Data	
Out of Limits	0101 0001
Unsettled	0101 0010
Out of Control	0101 0100
Undercurrent	0101 0100
Overcurrent	0101 0101
Error in Input Data	0101 0110

Note that the most significant bit is set if the 2724A is in REMOTE.

SECTION V - THEORY OF OPERATION

5.1 GENERAL

The information is this section, combined with that in sections VI and IX is all that is required to perform successful maintenance and calibration of the 2724A. Should further information be required then the user is referred to his nearest Valhalla Service Center for advice.

This section of the manual describes the operation of the 2724A. The functional description of 5.2 is intended to assist the user in gaining a general understanding of the principals behind the operation of the 2724A referring to the block diagram shown in Figure 5-1. Following this functional description in 5.3 are detailed descriptions of each circuit block referring to the schematic diagrams in section IX. These circuit descriptions are intended to provide the user with sufficient information on circuit operation to permit efficient troubleshooting.

5.2 FUNCTIONAL DESCRIPTION

The "active ohm" technique used in the 2724A is a unique analog circuit providing stability and high accuracy for a wide range of values and conditions.

Referring to the block diagram in Figure 5-1, the current (I) from the measuring device flows into the I HI terminal of the 2724A, through the reference resistor (Rref), into the internal ØV of the 2724A returning via the power rails from the output of the variable gain amplifier to the I LO terminal. Thus, the voltage across the reference resistor is I*Rref (and thus the voltage at the V HI and I HI terminals).

With the variable gain amplifier at a gain of K (Ø<K<1) the voltage at the I LO terminal (and thus the V LO terminal) is K*I*Rref. Therefore, it can be seen that the voltage between the V HI and LO terminals is given by I*Rref*(1-K), thus simulating a resistor of value Rref*(1-K). The digital circuitry in the 2724A is present to control I/O operations, set the required Rref and K, and to check the exact value of K (correcting any errors present) by utilizing the A-to-D convertor.

5.3 DETAILED CIRCUIT DESCRIPTIONS

5.3.1 REFERENCE RESISTORS

The resistors R121, 122, 123, 124, 125 and 126 are the reference resistors used in 2724A. As can be seen in schematic

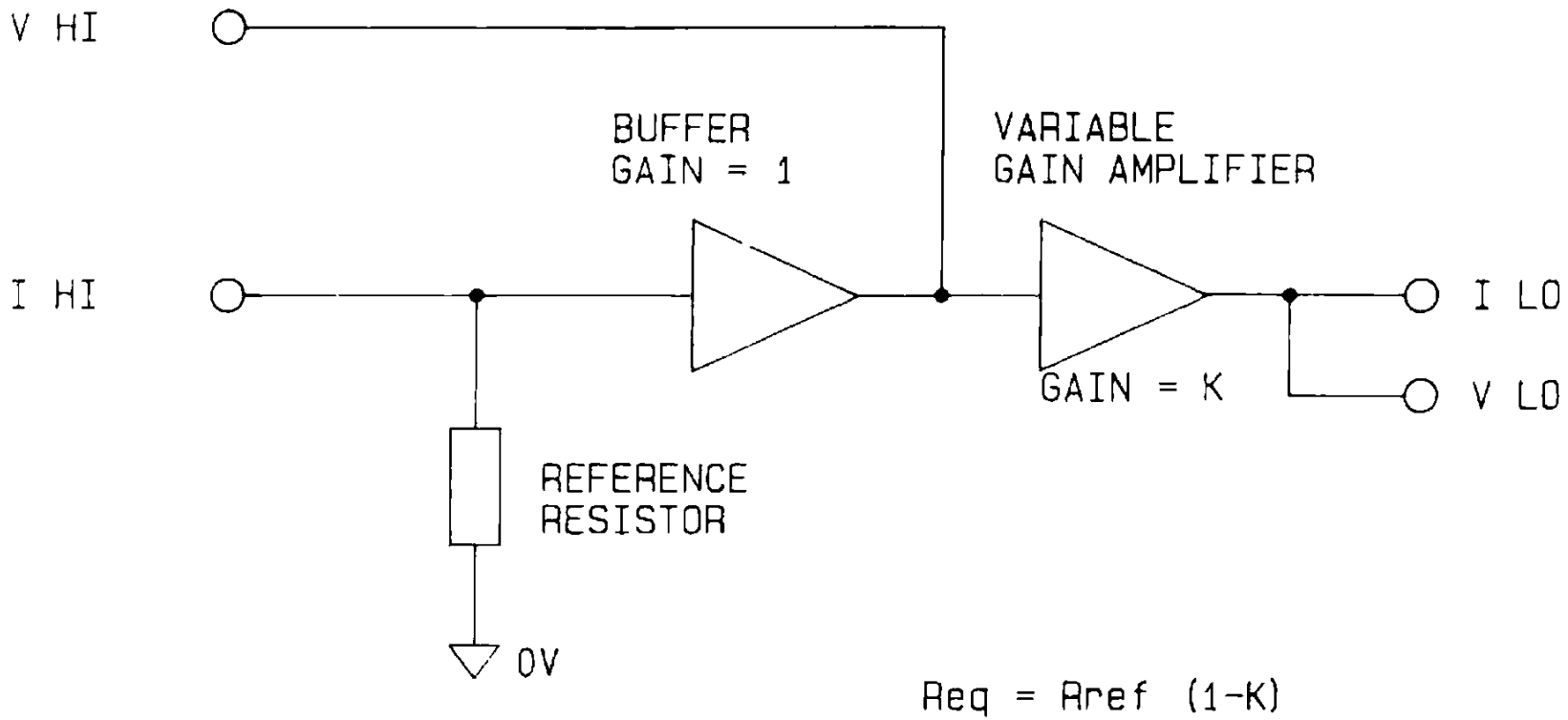


FIGURE 5-1 ACTIVE OHM BLOCK DIAGRAM

2724-Ø7Ø sheet 1, the actual reference resistor in use is selected by the relays RLA, B, C, D, E and F the actual selections are shown below:-

VALUE	RLA	RLB	RLC	RLD	RLE	RLF	REFERENCE RESISTOR
Ø to 12Ø ohm	Off	On	Off	Off	On	On	R124 (1ØØohm)
Ø.12 to 1.2 Kohm	Off	On	Off	On	Off	Off	R124+123 (1Kohm)
1.2 to 12 Kohm	Off	On	On	Off	Off	Off	R124+123+122 (1ØKohm)
12 to 12Ø Kohm	On	On	Off	Off	Off	Off	R124+123+122+121 (1ØØKohm)
Ø.12 to 1.2 Mohm	Off	Off	Off	Off	On	On	R126 (1Mohm)
1.2 to 12 Mohm	Off	Off	Off	Off	Off	Off	R126+125 (1ØMohm)
12 to 11ØØØMohm	Off	Off	*	*	Off	Off	R126+125 (1ØMohm)

* See 5.3.2.2 for action of RLC and D above 12Mohm.

5.3.2 REF BUFFER

5.3.2.1 Unit Gain Buffer

The unity gain buffer is formed by the amplifier IC101, powered by rails (+ and - 5 volts) that follow the input as formed by IC104, 105 and 106.

IC104 simply buffers the voltage developed across the reference resistor and forces the +5V (from IC105) and -5V (from IC106) rails for IC101 to follow this voltage. This amplifier's output is also used to detect the overcurrent situation by means of D103, 104 conducting when the voltage exceeds 13.7V, thus turning on the LED part of the opto-isolator IC411 (informing the MPU of the overcurrent).

IC101 is also connected as a unity gain, non-inverting buffer, driving the V HI terminal, reference resistor section guarding and also the variable gain amplifier (for values below 12Mohm). The input offset and input leakage current of this section are corrected for by the adjustment of RV103 and 102 respectively.

At the input to this section is a selectable filter formed by RLL, C112 and C111. This filter is selected by the Slow or Fast mode of the 2724A as defined by the user. In slow mode RLL is on, thus using C111 as the filter, whereas in fast mode RLL is off, thus using C112 as the filter.

At the output of this section is an adjustment (switched on by RLG when the Ohms Multiplier is not in use) RV101 which is used to compensate for thermal emfs on RLG and R101.

5.3.2.2 Ohms Multiplier

In order to avoid the necessity of reference resistors above 10Mohm, the effective value of the reference resistor is multiplied by 10, 100 or 1000 by the Ohms Multiplier. This is formed by the amplifier comprised of IC102 and 103, the feedback resistor network R114 and the relays RLG, H and J.

For values below 12Mohm this circuit is bypassed by RLG being off. For values between 12 and 120Mohm this amplifier is set to a gain of 10 by RLG being on, RLH being on and RLJ being off. Note that RLH and RLC are driven together as are RLJ and RLD. For values between 120 and 1200Mohm this amplifier is set to a gain of 100 by RLG being on, RLH being off and RLJ being on. For values between 1200 and 11000Mohm this amplifier is set to a gain of 1000 by RLG being on, RLH being off and RLJ being off.

The amplifier formed by IC102 and 103 is a two-stage amplifier providing the offset voltage characteristics of IC103 combined with the output swing capability of IC102.

The output of IC102 is also used to detect overcurrent by means of D101 and D102 conducting when the output exceeds 13.7V thus turning on the LED portion of IC411 in a similar manner to that described in 5.3.2.1 above.

5.3.3 VARIABLE GAIN AMPLIFIER

This section provides a variable gain from Ø to unity. It comprises of two sections:-

5.3.3.1 MDAC's

There are two virtually identical MDACs (Multiplying D to A Convertor) in the 2724A. These are formed by IC201 and 202 (Most Significant DAC) and IC203 and 204 (Least Significant DAC). The only difference between these is the accuracy of the DAC device used.

Each MDAC is a 12-bit (i.e. 1 in 4096 resolution) device using the output of the REF BUFFER as its reference voltage input. The output of each MDAC is buffered and converted into a voltage by IC202 or 204. The gain of each MDAC is adjusted by the digital codes present on the inputs D1 (Least significant) to D12 (Most significant), if this is considered as a number K (between Ø and 4Ø95) then the output of IC202 (or 204) is given by K*Vref/4096 (where Vref) is the voltage on the reference input).

The MDACs are powered by an 8V rail provided from the +18V rail by IC206.

5.3.3.2 Output Buffer

The output buffer is an inverting buffer summing the outputs of the two MDAC's with a 1ØØØ:1 weighting as defined by R202:R201.

When the input current is less than 1ØmA (approximately) then the amplifier IC205 is used directly to drive the V LO and I LO terminals, otherwise a high-power buffer amplifier IC207 is inserted into the output of IC205 to sink (or source) the required current.

The output of the Output Buffer is protected by the two 15V zener diodes D203 and 204.

If the 2724A has OPTION CPR fitted then IC207 is deleted and the NO (Normally Open) contact of RLK is directly taken to Øv. RLK is now used to select CPR mode of operation, shorting V LO and I LO to Øv, thus the output value is that of the selected reference resistor. With RLK off the 2724A operates normally, however it may only provide 15mA of current.

The actual output of the Variable Gain Amplifier is monitored by the A-to-D convertor through R210 (filtered by C203) in order to monitor the actual gain.

Under overcurrent condition the MPU pulses TR201 on to force the output to a high voltage in order to ensure that auto-ranging DVM's which require 2ØV (or more) to up-range will do so.

5.3.4 A/D CONVERTOR

The A/D convertor is shown on schematic 2724-070 sheet 3 and comprises four sections:-

5.3.4.1 Pre-Scaling and Input Multiplexing

The input to the A-to-D convertor is selected from one of three sources (Zero, REF BUFFER output, V LO terminal) by the analog multiplexer IC301. This signal is pre-scaled to within 4-13V by the amplifier IC302, gain defining resistors R301-308 and multiplexer IC303.

Control of both multiplexers is by the MPU under software control, the input being multiplexed in order to calculate the ratio between the output of the REF BUFFER and the voltage on the V LO terminal, eliminating zero errors.

5.3.4.2 Reference Voltage Generator

The reference voltage for the A-to-D convertor is generated by IC307 at 6.95V (nominally). The A/D convertor requires both positive and negative reference voltages thus the reference is inverted by amplifier IC308 to produce the required negative voltage. RLM selects the polarity used by the A/D convertor to operate for the polarity of input current to the 2724A.

5.3.4.3 FET Switch Drives

The A/D convertor requires two high-speed FET switch drivers. These are both essentially identical and are comprised of IC309 in combination with TR303, R319-321 and C304 (TR302 drive) or TR304, R326-328 and C305 (TR301 drive). These drivers take the TTL level signal (REF DRIVE) which is referred to the -18V rail, and provides the wide voltage swing required to switch the FETs TR301 and 302.

5.3.4.4 Current Balance

The currents formed through R309 (from the input), R317 (from the reference) and R318 (from the FET switches) are summed in the integrator IC304 and C302. Any imbalance in these currents will cause the output of IC304 to move thus causing the comparator IC305 to switch. The output of this comparator is synchronized to clock and fed back to drive the REF DRIVE line, thus causing the current through R318 to be altered. The 667Hz input to the A/D convertor through C301 and R310 forces the convertor to operate continuously at a frequency which is independent of the input. This synchronization is performed in the MPU section described in section 5.3.5.1.

Under normal operating conditions the output of IC304 will be a distorted triangle wave and the output of IC305 will be a TTL level square wave with a duty cycle proportional to the input voltage at that time. The waveform at the junction of TR301, TR302 and R318 will also be a square wave with levels of ØV and Vref (Vref will be -6.95V with a positive input, +6.95V with a negative input).

5.3.5 FLOATING SECTION ISOLATION

All of the section described above are floating, being driven to the common-mode voltage of the measuring device. All of the I/O and MPU circuitry in the 2724A are grounded thus isolation is required between these two parts of the 2724A. This is achieved by the circuitry shown in schematic 2724-070 sheet 4. The circuitry used is basically simple employing opto-isolators IC's 401 to 413 to isolate all of the signals passing between the two parts. IC414 and TR's 403 to 406 are used to drive the relays from the outputs of the opto-isolators. TR401 is used to drive the 667Hz drive to the A/D convertor. TR402 is used to drive TR201 in the Variable Gain Amplifier. The various

discrete and film-network resistors being used for current definition in the LED's in the opto-isolators and as pull-ups for the opto-isolator transistor outputs.

5.3.6 MICROPROCESSOR

The microprocessor section of the 2724A is shown in schematic 2724-070 sheet 5 and is in five sections:-

5.3.6.1 A/D Convertor Logic

The logic for synchronization of the current balance output to the main clock and the accumulation of A/D convertor measurements is performed by ICs 501 to 509 in combination with the MPU under software control. The opto-isolated output of the current balance is first passed through an exclusive OR gate in IC503 which is used to invert or non-invert this signal dependent on the polarity of input current (and thus reference used in the A/D convertor), this signal is now synchronized with the 12.8MHz clock and de-bounced by ICs 501, 502 and parts of ICs 503 and 504. The true output of the final stage (IC502) is used as the REF DRIVE signal for the FET switch drivers in the A/D convertor (after isolation) and the untrue output is gated with the clock to generate a count signal for the 16-bit counter formed by IC506 and 507. Due to the already described operation of the A/D convertor this signal will be a group of clock pulses every 1.333 milliseconds, with the number of pulses being proportional to the input voltage to the A/D convertor at that time. IC505 is used to produce a pair of pulses immediately following each of these groups of pulses, the first pulse is used to latch the count in the 16-bit counter into latches IC508 and 509, the second being used to clear the 16-bit counter to zero ready for the next group of pulses. Thus, at 667Hz, the latches IC508 and 509 are updated with the latest 16-bit measurement taken by the A/D convertor. The second pulse from IC505 is also used to interrupt the microprocessor, informing it that a new measurement may be read from the latches. The microprocessor then accumulates these measurements until sufficient resolution is obtained.

5.3.6.2 Opto-Isolator Latches/Drivers

The opto-isolators are driven by several latches IC514-517 and 526 with the required data. The microprocessor loads the data into these latches by simply "writing" the 8-bit data into them, each being at a defined address.

5.3.6.3 Microprocessor (MPU)

The microprocessor is a Motorola MC6809 8-bit device running from a 6.4MHz clock generated from the main 12.8MHz clock. It is beyond the scope of this manual to fully describe the operation of the 6809 and thus the user is referred to the many data sheets and books available on this device.

The connections to the microprocessor fall into four groups or busses:

1) Address Bus - This 16-bit parallel bus contains the address of the device or byte to be read from/written into. It is not used as a tri-state bus nor is the MPU ever halted, thus these bus signals should always be at the two logic levels (except during transition).

2) Data Bus - This 8-bit parallel bus contains the data read from/written to the device whose address is present on the address bus. This is a tri-state bus and various different levels will be seen on these signals.

   The MOS devices on this bus are unable to drive the many TTL devices required on this bus, thus they are buffered from the MPU data bus by IC511.

3) Control Bus - This describes the two lines Ø and R.

   Ø is the MPU clock and is 1/4 of the 6.4MHz clock to the MPU (i.e. 1.6MHz). All buss transactions take place at the trailing (i.e. 1 to Ø) edge of this signal.

   R is the read/write control signal and is a 1 whilst the MPU is reading data and is a Ø whilst writting. Neither of these two signals are tri-state and thus are always at one of the two logic levels.

Interrupts - There are four interrupts used in the 2724A.

RESET - This line is normally pulled to +5V by R503, but is held low for approximately 300mS (by C503) after applying power to the 2724A. This action causes initialization of the MPU and restarts the software.

NMI - This line is normally at a logic 1, going to a Ø when a new measurement is available from the A/D convertor.

FIRQ - This line is normally at a logic 1 and is driven from the display drive circuitry to indicate that the next position in the display is to be updated.

IRQ - This line is normally at +5V, being pulled-up by R504. It is used by the IEEE-488 interface to indicate that an IEEE bus action has taken place.

5.3.6.4 Address Decode

The decoder IC's 524 and 525 perform the decoding of the address bus into the individual addresses of the various devices connected to the data bus. Only the lines A9 to 15 are used and the decode is synchronized to the MPU clock by IC708.

5.3.6.5 Memory

The memory in the 2724A comprises ROM (IC519 - 8kx8 UVEROM), static RAM (IC's 520 and 521 - 1kx4 each, organized as 1kx8) and non-volatile RAM (IC's 522 and 523 - 256x4 each, organized as 256x8). Each of these has its address range decoded by IC524. The non-volatile memory also has address decode for addresses controlling the reading and writting of the NOVRAM portions of these memories. IC522 is used to hold the calibration data for the 2724A and thus has its write control switched by the key switch whilst IC523 is used to hold the user memory data and thus does not.

5.3.7 CLOCK, DISPLAY AND KEYBOARD

IC527 and Y501 form the 12.8007 MHz oscillator for the main system clock. This clock is then divided down in sequence by IC528 (giving 128KHz), IC529 (giving 667Hz). These components are shown on schematic 2724-070 sheet 6.

The display is a 16-digit 14-segment vacuum fluorescent type, multiplexed at 2.66KHz by the main system clock. The display itself is mounted on the Display PCB behind the front-panel and is shown on schematic 2724-071. The grid of each decade is pulled up by ICs 703 and 704 to +37V one at a time, as defined by decoder IC705 and inverters IC706 and 707. Each time that a new decade is turned on the latch IC532 is set by the clock thus blanking the display and interrupting the MPU. The MPU responds by writting the new data for that decade into latches IC512 and 513, driving the drivers IC701 and 702 with the new data for each individual segment of the new display decade. The MPU then clears the latch IC532 enabling the display again. The front-panel LEDs DS702-708 are driven by the spare output of the 16-bit latch driving the segments of the display, buffered by TR701. The actual position of the decade in the display is synchronized by the last decade drive being fed-back into the MPU via buffer IC510.

The keyboard formed by the front-panel switches shown on schematic 2724-071 are also multiplexed along with the display. Each decade of the display causes a group of four switches to

be enabled. The MPU reads the switches by reading the buffer IC510.

5.3.8 IEEE-488 INTERFACE

This option comprises a PCB mounted on the rear-panel of the 2724A. The circuitry contained on this PCB is shown on schematic 2724-072.

ICs 3 and 4 buffer the actual IEEE bus lines from the IEEE interface IC2, which is buffered from the MPU data bus by IC1.

An exact description of the operation of the IEEE-488 bus and that of the IEEE interface IC2 are beyond the scope of this manual and the user is referred to the IEEE-488 (1981) standard and the Motorola documentation on the MC68488 chip.

5.3.9 POWER SUPPLIES

The circuitry providing the various power rails for the 2724A is shown on schematic 2724-070 sheet 7. This contains the transformer T1 whose outputs are rectified by D601-606, filtered by C601, 602, 604-607, regulated by ICs 601 to 606.

SECTION VI - CALIBRATION AND MAINTENANCE

6.1 A BRIEF DISCUSSION OF THE CALIBRATION PROCEDURE

The calibration of the 2724A is done using a current source, a standard resistor and a voltmeter. With the switch box you can switch the voltmeter and the current source back and forth between the 2724A and the standard resistor. Using ohms law (R*I=E) you can tell what the voltage would be. For example 100 ohms at 10 milli Amps = 1 volt. When the 2724A asks you for 100 ohms at 10 milli Amps you need to put in exactly 10 milli Amps. The way to obtain exactly 10 milli Amps is to refer to the standard resistor.

Switch the switch box to the standard resistor. Now, using ohms law, the value of the standard resistor is 99.9985 ohms, the current 10 milli Amps, so the voltage would be 0.999985 volts. If the voltmeter does not read that voltage, then adjust the current source until it does. When you have it adjusted, switch the switch box back to the 2724A. You should read 1.00000 volts on the voltmeter. If you don't, then adjust the 2724A using step control until you do. Switch back to the standard resistor you should read the value of the resistor 0.999985, switch back to the 2724A you should read the value of 2724A 100 ohms 1.00000 volts.

Most of the cal steps are done this way, however, there are a few that require more explanation. 0 ohms at 10 milli Amps gain ohms law 0*10 milli Amps=0.00000 volts. If it's not 0 volts, but a negative voltage you can use the step control to step it up to 0 volts. If the voltage is positive you have to enter in the value on the 2724A key board. Let's say the voltage was 0.00543 volts using ohms law that would be 0.543 ohms. So type .543 and hit the ohm key on the 2724A. The 0 ohms at 10 milli Amps will appear twice. The second time you see it, make sure you move the voltage sense lead over to the I terminals, so both leads are coming off the I terminals. This is a two wire 0 ohm calibration. After this calibration point, make sure you move your voltage sense leads back to the V terminals.

The next step of concern is when it asks you to apply 10 milli Amps you must apply 10 milli Amps. It will ask you twice. Watch the voltmeter. The first time it will be around 0 volts. The next time it will be around 10 volts. If your unit is a CPR 2724A then it will ask you to calibrate the internal resistors in the 2724A 100 ohms, 1K ohm, 10K ohms, 100K ohms, 1M ohm, and 10M ohms. For each of these cardinal point resistors you will use the switch box, the current source, the standard resistor, and the voltmeter in the same way you did before except instead of adjusting the 2724A with the step control for 1.000000 volts, you need to enter in the value. For instance CAL 100 ohm CPR, you use the switch box switch to your standard resistor and set your

current to 10 milli Amps. Now switch to the 2724A, say the voltmeter reeds 1.000056 volts, you would enter on the 2724A keyboard 100.0056 and hit the ohms button.

You use this same procedure for all the other cardinal point resistors. Make sure you hit the ohm key when it asks for ohms, the K ohm key when it asks for K ohms, and the M ohm key when it asks for M ohms. Any cal step can be skipped except apply 10 milli amps. If you are going to skip steps remove any current input while skipping, except when it asks you to apply 10 milli Amps do so. When skipping over CPR cal make sure you press ohm for ohm, K ohm for K ohm, and M ohm for M ohm. If all goes well the 2724A will say Fast Data OK , Cal Data OK , Turn Cal Switch Off.

6.2 GENERAL

This section provides calibration procedures using the front panel controls or the IEEE-488 bus. A troubleshooting chart is also provided which lists a number of trouble symptoms and the components or circuit areas which may produce them. It should be noted that a Calibration Kit (Option M-24) containing a switch assembly, cables and miscellaneous items is available from Valhalla.

6.3 CALIBRATION

It is recommended that calibration be performed with the 2724A fully assembled (i.e., with covers in place) and after power has been applied for a minimum of one hour to achieve temperature stabilization. Calibration may be accomplished manually, using the front panel controls, or automatically through the IEEE-488 interface. In all cases ACTIVE mode is used unless otherwise stated.

6.3.1 STEP 1 (NON-CPR UNITS)

The following procedure should be followed prior to full calibrations:

a) With a DVM with 1 micro Volt sensitivity monitor the voltage between V LO and V HI terminals. Disconnect any connections to the I terminals and select 100 ohms on the front panel. Adjust RV201 for A DVM reading of 0 +/- 30 micro Volts.

6.3.2 STEP 2 (NON-CPR UNITS)

These adjustments are required only if a component has been changed - periodic adjustment is unnecessary.

a) Ensure there are no connections to either front or rear terminal sets.

With a DVM with 1 micro Volt sensitivity monitor the voltage between the rear end of R126 and the rear end of R106 (use copper alligator clips).

Select SLOW-MODE AND FOUR-WIRE

Setting	DVM Reading	Adjust
0 ohm	0 ± 2 micro Volt	RV103
10 M ohm	0 ± 20 micro Volt	RV102

Repeat until no further adjustment is required.

b) Monitor the voltage between the rear end of R126 and the ground-plane near IC301. Adjust RV301 for a DVM reading of 0 ± 2 micro Volts.

c) Connect a current source to the "I" terminals and monitor the "V" terminals with the DVM.

Select 0 ohms Adjust RV101 such that the DVM reading at 500 micro Amps is 1/20th of that at 10mA.

2724A - Select 100 ohm Adjust RV103 such that the DVM reading at 500 micro Amps is 1/20th of that at 10mA.

2724A/CPR - Select CPR mode. Adjust RV103 such that the DVM reading at zero current is 0 ± 1 micro Volt.

6.3.3 2724CPR ADJUSTMENT PROCEDURE

This procedure replaces that contained in paragraphs 6.2.1 and 6.2.2 of the manual and concerns the procedure by which the internal adjustments in the 2724CPR are performed, the procedure for non-CPR units is unchanged. Note that all of these adjustments are recommended to be performed before each calibration of the unit. The user should note that the internal adjustments MUST BE MADE WHILE THE UNIT IS IN THERMAL EQUILIBRIUM. THUS THE TOP COVER MUST BE IN PLACE WHEN MEASUREMENTS ARE MADE. The user is recommended to use a top cover with small holes in it for the internal adjustments, this is available from Valhalla under the part number M24 (a switch box for the switching of the current source between the standard and the 2724 is also provided). If the top cover is lifted to perform the adjustment then the user

must ensure that the unit has thermally recovered before making any further measurements.

Equipment Required

a) A Digital Voltmeter with at least 1uV resolution at 1V and below, and 10uV resolution at 1 to 10V levels.

b) A current source providing 10mA, 1mA with less than 1ppm low frequency noise and better than 1V compliance capability.

STEP 1 - RV201 - ACTIVE/FAST Mode Output Voltage Offset

Monitor the voltage between the front panel V LO and V HI terminals with the DVM. Disconnect any connections from the I terminals and select 0 OHM/ACTIVE/FAST on the 2724CPR.

Adjust RV201 for a DVM reading of 0 +/- 50uV.

STEP 2 - RV103 - Current Sense Voltage Offset

Monitor the voltage between the front panel V LO and V HI terminals with the DVM. Disconnect any connections from the I terminals and select 100 OHM/CPR/SLOW on the 2724CPR.

Adjust RV103 for a DVM reading of 0 +/- 1uV.

STEP 3 - RV102 - Current Sense Current Offset

Monitor the voltage between the front panel V LO and V HI terminals with the DVM. Disconnect any connections from the I terminals and select 10 MEGOHM/CPR/SLOW on the 2724CPR.

Adjust RV102 for a DVM reading of 0 +/- 20uV.

STEP 4 - RV101 - Active Ohms Voltage Zero Offset

Monitor the voltage between the front panel V LO and V HI terminals with the DVM. Select 0 OHM/ACTIVE/SLOW from the 2724CPR. Alternately connect the source of 10mA and 1mA to the I terminals. The accuracy of the current source should be better than +/- 0.5% at each level.

Adjust RV101 such that the DVM reading at 10mA is ten times (i.e. the same resistance value) the DVM reading when 1mA is applied.

STEP 5 - RV301 - ACTIVE/CPR Alignment

Monitor the voltage between the front panel V LO and V HI terminals with the DVM.

i) Select 100 OHM/CPR/SLOW from the 2724CPR. Apply the source of 10mA current to the I terminals of the 2724CPR. Noting the DVM reading in CPR mode, select the ACTIVE mode (press the CARD PNT key). Note the change in DVM reading (in uV) between the two modes of operation (ensure sufficient time is given for the 2724CPR to settle before noting each DVM reading).

ii) Repeat the measurements taken in i) above with 1mA current instead of 10mA. Adjust RV301 such that the change in DVM reading at 1mA is 1/10th that at 10mA (within 1uV) repeating i) and ii) as required until no further adjustment is required.

All of the above steps should be repeated until no further adjustments are required. The 2724 is now ready for front panel calibration of the values. In cases where a component has been changed, or where the 2724 was found to have excessive errors in any step, then it is recommended that the entire calibration procedure be repeated.

6.3.4 FRONT PANEL CALIBRATION

This section shows the procedure required for the 2724A to "Learn" its errors, via the front panel.

The general concept behind the procedure for the 2724A is to adjust the output to be that which it should be, then ENTER this adjustment. The 2724A now remembers the amount of adjustment required and will automatically use this for all future outputs required.

It should be noted that:

a) A calibration step in 6.3.1 and 6.3.2 may be skipped by pressing either the ohm, K ohm or M ohm key without adjusting the output.

b) THE 2724A MUST NOT HAVE POWER REMOVED WHEN THE KEY OPERATED SWITCH IS IN THE "CALIBRATE" POSITION.

c) If errors of greater than 1% (or 10 milli ohms) are found, then the user should carefully check his calibration equipment before adjusting the 2724A.

6.3.5 STEP 3 - ACTIVE MODE CALIBRATION (all units)

A Equipment Required

1. Current Source capable of delivering 1 micro Amp to 10 milli Amp @ 10V compliance and 30mA @ 3V compliance. Current noise to be less than ± 2 ppm peak. (e.g. Valhalla 2701C +n 2500 in combination)

2. Resistance Standards of 100 ohm, 1 K ohm, 10K ohm, 100K ohm, 1M ohm and 10M ohm with values known to within 10 ppm.

3. Digital Voltmeter with at least 10 micro Volt resolution @ 10v or 3v, and 1 micro volt resolution @ 1v or below.

4. Cabling or switch box - See Figure 6-1. (Switch box is available from Valhalla under Option M-24. Cabling must be low thermal and have a leakage resistance of 10^{13} ohms.

B Calibration of the resistance values requires that the key operated switch in the rear panel of the 2724A be placed in CAL position.

C After performing step B (above) the unit will display "CAL O OHM/030mA (010mA if CPR option unit).

The user should apply the requested current to the I terminals. The current may be within ± 10% of the requested value.

If the unit exhibits a positive error, then the user should use step control to zero (within 1 milli ohm) the output and, staying in step control mode, press the ohm, K ohm and M ohm key. The unit will step on to D, below.

D The display will now show "CAL 100 OHM/030mA" (010mA if option CPR unit).

The user should adjust, (using a standard resistor of the same value as the 2724A, and DVM) the current source to within ± 5 ppm of the requested value. Now connect the current to the I terminals of the 2724A and monitor the V terminals with the DVM.

Adjust, using step control, the 2724A's value until the exact cardinal value is achieved. It is suggested that

the current source be rechecked as shown above, and then the 2724A reset until no further adjustment of the 2724A is required in order to obtain the required value to within ± 5 ppm.

Ensuring that the 2724A is still in step control mode, and that the current is flowing in the I terminals, press the ohm, K ohm or M ohm key. The unit will then step on to the next step.

E The display will now "CAL 1KOHM/010mA"

The user should now follow the same procedure as shown in D Above.

F The display will now show "CAL 10KOHM/1mA"

The user should now follow the procedure shown in D above.

G The display will now show "CAL 100KOHM/100µA"

The user should now follow the procedure shown in above.

H The display will now show "CAL 1MOHM/10µA"

The user should now follow the procedure shown in D above.

I The display will now show "CAL 10MOHM/1µA"

The user should now follow the procedure shown in above.

J The display will now show "CAL 100KOHM/1µA"

The user should now follow the procedure shown in D above, however it is only necessary to adjust the current source and 2724A to within ± 0.05% and not ± 5ppm as used in D.

The display will now show "CAL 0OHM/30mA" (10mA if CPR unit)

The user should note that the TWO WIRE LED is on during this step, and a two-wire check should be made (monitor I instead of V terminals) in a similar manner to that in step C above.

The display will now show "APPLY 10mA" (Note: Units with early ROM's, before V-FB will display 100mA)

The user should apply the requested current (+ 10%) to the I terminals and monitor the V terminals with the DVM.

The DVM reading will settle to approximately zero.

When settled press the Ω , KΩ or MΩ key. The unit will step on to the next step.

The display will now show "APPLY 10mA"

The same procedure as in L, above, should be followed, the DVM reading settling to 10 volts instead of zero.

6.3.6 STEP 4 - CPR CALIBRATION (Option CPR Only)

A The display will now show "CPR 100Ω/CAL"

The user should adjust (using the standard resistor of the same value as the nominal 2724A value, and the DVM) the current source to provide 30mA ± 0.3μA.

Now connect the current source to the I terminals of the 2724A and monitor the V terminals with the DVM.

The user should now enter the actual value of the 2724A - ENSURE THAT THE ENTRY IS FULLY CORRECT BEFORE PRESSING A UNITS KEY (Ω, KΩ or MΩ) AND THEN ENSURE THAT THE CORRECT UNITS ARE ENTERED.

After pressing the Ω, KΩ or MΩ key above, the unit will step on to the next step.

B The display will now show "CPR 1KΩ/CAL"

The procedure in A above should be followed with 10mA ± 5ppm current.

C The display will now show "CPR 10KΩ/CAL"

The procedure in A above should be followed with 1mA ± 5 ppm current.

D The display will now show "CPR 100KΩ/CAL"

The procedure in A above should be followed with 100μ A ± 5ppm current.

E The display will now show "CPR 1MΩ/CAL"

The procedure in A above should be followed with 10μA ± 5ppm current.

F The display will now show "CPR 10MΩ/CAL"

The procedure in A above should be followed with 1μA ± 5 ppm current.

6.4 STEP 5 - COMPLETION OF CALIBRATION

After step 3 (2724A) or step 4 (CPR option) the unit will report on the calibration.

If step 3 L or M was not completed (insufficient current) then "NO FAST DATA" will be displayed and the FAST mode calibration data left as it was before calibration.

If both step 3 L and M were successfully completed then "FAST DATA OK" will be displayed and the FAST mode calibration data will be updated.

If any calibration data is outside the tolerance limits of the 2724A then "OUT OF CAL" is then displayed, otherwise "CAL DATA OK" is displayed.

The unit will then display "TURN CAL SW OFF". The user should now turn the rear panel key operated switch to the OPERATE position. Calibration is now completed.

6.5 AUTOMATIC CALIBRATION

The procedure for automatic calibration utilizing the IEEE-488 bus is the same as for the front panel calibration. DIGIT STEP CONTROL or direct entry methods of calibration may be used. The messages required are the normal ones used for STEP CONTROL or numeric entry. The present calibration point may be easily ascertained by simply reading the 2724A display.

6.6 TROUBLESHOOTING

It is not possible to anticipate all failure modes of the integrated circuit devices and other components of the Model 2724A. Therefore, the servicing technician should be familiar with the contents of Section V Theory of Operation. Knowledge of circuit operation is a requisite for efficient servicing of the 2724A. As a servicing aid, the following troubleshooting chart, Table 6-1, lists a number of fault symptoms and possible sources. If the exhibited symptom is not listed or a check of the sources listed does not locate the defective component, it will be necessary to employ normal troubleshooting procedures. In general, the list of potential error sources are in order of probability (most probable first).

Figure 6-1. Calibration connections

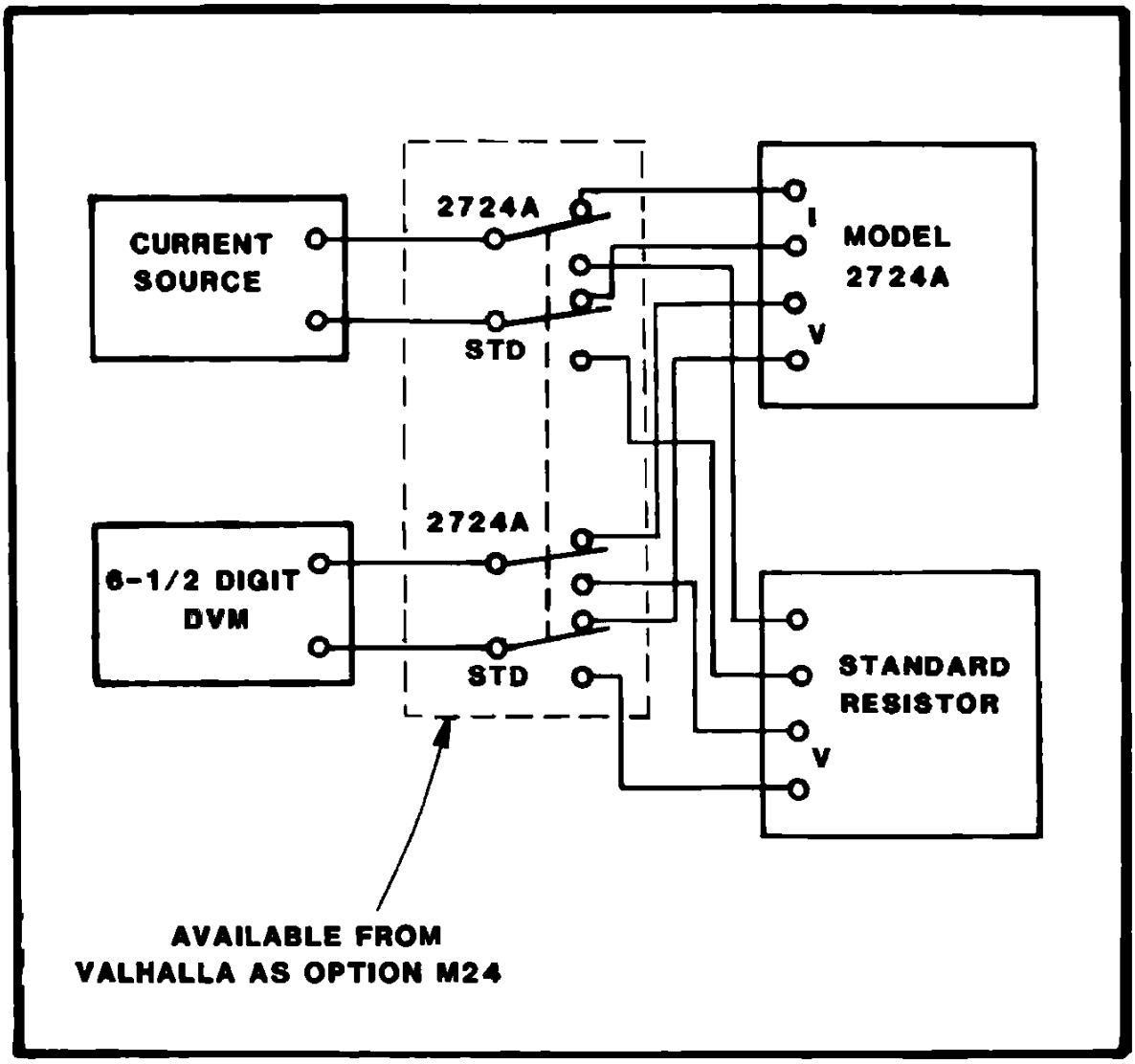


Table 6-1

Troubleshooting Guide

SYMPTOM	POSSIBLE FAULT AREA
All decimal points displayed.	MPU or ROM failure. Address Decode.
All middle segments displayed.	RAM failure.
"OUT OF CAL" displayed.	Unit mis-calibrated. Reference resistor bad. MDAC bad.
"CAL DATA BAD" displayed.	Bad NOVRAM.
"CLOCK FAILURE" displayed.	Incorrect display multiplex frequency.
"CANNOT CONTROL" displayed.	Unit mis-calibrated. MDAC bad. A/D Convertor bad.
"MEMORY DATA BAD" displayed.	If repeats after turning power off and back on, then NOVRAM bad.
"A TO D TIMEOUT" displayed.	A/D Convertor bad. Unit mis-calibrated. Unit in overcurrent, but overcurrent circuitry bad.
No display or LEDs.	Faulty line power or fuse. Power Supply bad. Clock bad. Display circuitry bad. MPU bad. Address decode bad.
Large errors, all ranges.	Check measuring equipment. A/D convertor bad. REF Buffer bad. RLB bad.
Large error, one range.	Reference resistor bad. Relay bad. Unit mis-calibrated.

Table 6-1 (Continued)

Troubleshooting Guide

SYMPTOM	POSSIBLE FAULT AREA
Error, increasing with higher value.	Excessive Leakage in relays. REF buffer bad.
Noise, all ranges.	A/D convertor bad. REF buffer bad. Check measuring equipment.
Noise, particularly on higher values.	Reverse leads to measuring equipment. REF buffer bad.
Output stuck at high (>10v) Voltage, with no input	Variable Gain Amplifier.
Value "hunts" between two or three levels.	MDAC bad.

SECTION VII - AVAILABLE OPTIONS

7.1 GENERAL

This section describes several options available from Valhalla Scientific to increase the utility of the Model 2724A.

7.2 OPTION TL-1

Option TL-1 is a Talk/Listen IEEE-488 (GPIB) Interface. It is fully described in Section IV.

7.3 OPTION RX-3

Option RX-3 provides all parts required for mounting the Model 2724A in a 19-inch equipment rack.

7.4 OPTION BBL

Option BBL is a dual shielded cable equipped with banana plugs. It is 48 inches long with dual banana plugs of the highest quality and low leakage.

7.5 OPTIONS GP1, GP2

Options GP1 and GP2 are IEEE-488 (GPIB) cables of one meter and two meters in length, respectively.

7.6 CALIBRATION KIT M24

This option contains a switch assembly, cables and miscellaneous items for aiding calibration of the Model 2724A.

7.7 OPTION CPR

This option enables the user to select the "CPR" mode of operation (see section III).

7.8 SP-2

This option provides a selection of the most probable components to fail during 24 months of operation.

SECTION VIII - USEFUL HINTS

8.1 GENERAL

This section contains a number of useful hints that will assist the user in obtaining maximum utility from the Model 2724A. As experience with the Model 2724A in the field is accumulated, this section will be expanded in later issues of this manual.

8.2 UNSTABLE TEST CURRENTS

As can be seen in the specifications, the Model 2724A requires approximately one second to respond fully to a change in test current. Thus, with ohmmeters having the facility of automatically switching polarity or to zero current (i.e., HP 3456), this facility must not be used.

With units having "uncontrollably" unstable test currents, then FAST mode or CPR mode (if fitted) should be used (e.g. Fluke 8502 series).

8.3 WHEATSTONE BRIDGE COMPARISON TO STANDARD RESISTORS

The Wheatstone bridge technique for comparing two four-terminal resistors will not operate with the Model 2724A. This is because all of the current must pass through the current (I) terminals of the unit. Thus, the lead compensation current would cause an error as shown in Figure 8-1. Using the system for comparing the Model 2724A wired as a two-terminal device presents no problems, but produces greater errors.

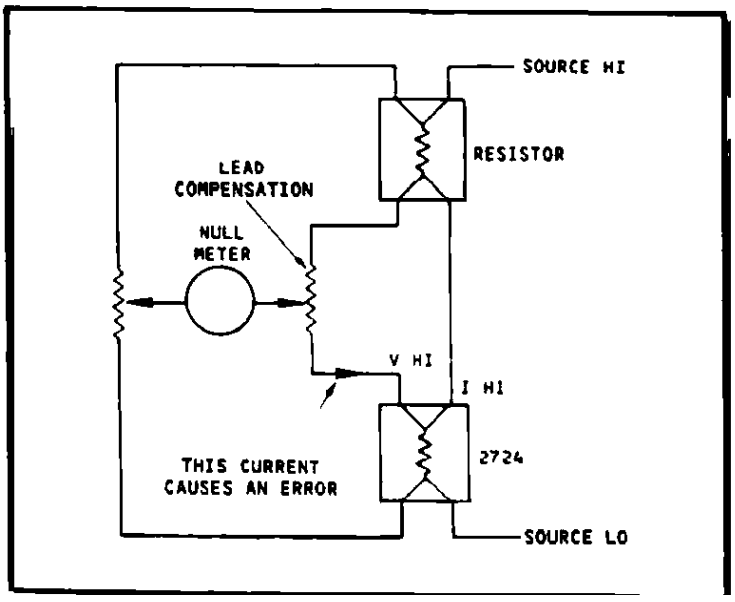


Figure 8-1. Wheatstone Bridge Error.

8.4 USE WITH HIGH RESISTANCE METERS

Several hints are provided here to help when using the Model 2724A with high resistance meters.

1. Remember that the maximum currents are:

   120M ohms -- 12Ø nA
   12ØØM ohms -- 12 nA
   12,ØØØM ohms -- 1.2 nA

   These correspond to 14 volts at full-scale.

2. Noise is significantly worse when used with a constant current source rather than a constant voltage.

3. Great care must be taken with regard to leakage. Regular PVC wire does not provide sufficient insulation. Teflon or polyethylene insulation should be used.

4. Care should be taken to minimize line pick-up. Outer shields of coaxial cables, LO terminals of meters, etc., MUST always be taken to low impedance nodes. If unstable indications are observed, try reversing the leads to the meter.

8.5 USE WITH ACTIVE FEEDBACK CURRENT SOURCES

Some ohmmeters utilize an active technique of ohms measurement and may not operate properly with the equivalent circuit of the 2724. In general the use of CPR mode (if fitted) or FAST mode is recommended. If this is not possible it has been found that a 0.1µF to 1µF capacitor (preferably Polycarbonate or Polystyrene) across the I terminals will usually prevent problems.

8.6 LINE INTERFERENCE

Some multimeters have poor line interference rejection when operated as an ohmmeter, thus care with line pick-up must be exercised. In extreme cases (e.g. when line cables run close to input cables) it may be found beneficial to place a capacitor across the I terminals to produce a 100 ms time constant. It may occasionally be beneficial to reverse the leads since some DMM's have better rejection on one terminal than the other (e.g. Guildline 9574 and Solartron 7060 should have cable shields taken to HI).

ADDITIONAL USEFUL HINTS

Rack mounting

The 2724 is sensitive to thermal gradients in the surrounding ambient, thus the user must ensure that at least 1.75" of free air is allowed around the 2724 when rack mounted. The user should also ensure that the unit immediately below and the unit above do not exhaust warm air onto the case of the 2724.

Voltage Offset Errors

Within the specification pages of this manual, values will be found for the maximum "Noise and Thermals" of the 2724. It will be noticed that only 4-wire operation is specified. The typical 2-wire thermal offsets are +/-20uV. This figure (and also the other figures quoted) will cause apparent errors on some ohmmeters. The user is recommended to ensure that the 2724 is in a thermally stable environment and that the leads between the 2724 and the meter under test are of reasonably good quality low thermal emf type in order to reduce this effect as far as possible. In severe cases the user may reverse the connections to the meter under test and take the average of the two ohmmeter readings, this will completely eliminate the thermal emf error, leaving only the effects caused by noise (typically < 1uV in CPR mode, < 3uV in ACTIVE mode).